

## SAFETY DATA SHEET

Creation Date 22-September-2009

Revision Date 27-March-2024

Revision Number 6

### 1. Identification

**Product Name** Cyanuric chloride

**Cat No. :** S36713

**CAS-No** 108-77-0

**Synonyms** 2,4,6-Trichloro-s-triazine; 2,4,6-Trichloro-1,3,5-triazine

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |              |
|---------------------------------------------------------|--------------|
| <b>Acute oral toxicity</b>                              | Category 4   |
| <b>Acute Inhalation Toxicity</b>                        | Category 2   |
| <b>Skin Corrosion/Irritation</b>                        | Category 1 B |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1   |
| <b>Skin Sensitization</b>                               | Category 1   |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3   |
| Target Organs - Respiratory system.                     |              |
| <b>Physical Hazards Not Otherwise Classified</b>        | Category 1   |
| Reacts violently with water                             |              |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Harmful if swallowed

Fatal if inhaled

Causes severe skin burns and eye damage

May cause an allergic skin reaction

May cause respiratory irritation

Reacts violently with water

**Precautionary Statements****Prevention**

Do not allow contact with water

Keep container tightly closed

Do not breathe dust/fumes/gas/mist/vapours/spray

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Use only outdoors or in a well-ventilated area

Contaminated work clothing should not be allowed out of the workplace

Wear protective gloves/protective clothing/eye protection/face protection

Wear respiratory protection

**Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER/doctor

Rinse mouth

Do NOT induce vomiting

Wash contaminated clothing before reuse

**Storage**

Store in a well-ventilated place. Keep container tightly closed

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                        | CAS-No   | Weight % |
|----------------------------------|----------|----------|
| 1,3,5-Triazine, 2,4,6-trichloro- | 108-77-0 | <=100    |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.                                                                                                                                                         |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## 5. Fire-fighting measures

|                                         |                                                                                                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Dry chemical. CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                           |
| <b>Flash Point</b>                      | > 200 °C                                                                                                           |
| <b>Method -</b>                         | No information available                                                                                           |
| <b>Autoignition Temperature</b>         | No information available                                                                                           |
| <b>Explosion Limits</b>                 |                                                                                                                    |
| <b>Upper</b>                            | No data available                                                                                                  |
| <b>Lower</b>                            | No data available                                                                                                  |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                           |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                           |

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Reacts violently with water.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 4             | 0                   | 2                  | W                       |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                     |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Evacuate personnel to safe areas. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid dust formation. |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                                        |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Do not expose spill to water.                                                                      |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe (dust, vapor, mist, gas). Avoid dust formation. Do not allow contact with water. Handle under an inert atmosphere. |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Storage.** Corrosives area. Keep refrigerated. Keep away from water or moist air. Store under an inert atmosphere. Protect from moisture. Keep container tightly closed in a dry and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong acids. Alcohols. Amines. Sulfides.

## 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures** Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection**  
**Hand Protection**

Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Butyl rubber   | recommendations   |                 |                        |
| Nitrile rubber |                   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                            |                               |
|----------------------------|-------------------------------|
| <b>Physical State</b>      | Powder Solid                  |
| <b>Appearance</b>          | White                         |
| <b>Odor</b>                | pungent                       |
| <b>Odor Threshold</b>      | No information available      |
| <b>pH</b>                  | 1 6.1 g/L aq.sol              |
| <b>Melting Point/Range</b> | 145 - 148 °C / 293 - 298.4 °F |

|                                        |                          |
|----------------------------------------|--------------------------|
| Boiling Point/Range                    | 190 °C / 374 °F          |
| Flash Point                            | > 200 °C                 |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | 0.025 hPa @ 20 °C        |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | 1.920                    |
| Solubility                             | Reacts with water        |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | C3 Cl3 N3                |
| Molecular Weight                       | 184.41                   |

## 10. Stability and reactivity

|                                  |                                                                                                                        |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                    |
| Stability                        | Moisture sensitive. Air sensitive.                                                                                     |
| Conditions to Avoid              | Temperatures above 40°C. Incompatible products. Exposure to moist air or water. Exposure to moisture. Exposure to air. |
| Incompatible Materials           | Strong oxidizing agents, Strong acids, Alcohols, Amines, Sulfides                                                      |
| Hazardous Decomposition Products | Nitrogen oxides (NOx), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen chloride gas                  |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                               |
| Hazardous Reactions              | None under normal processing. Reacts violently with water.                                                             |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Component Information

| Component                        | LD50 Oral                | LD50 Dermal               | LC50 Inhalation                |
|----------------------------------|--------------------------|---------------------------|--------------------------------|
| 1,3,5-Triazine, 2,4,6-trichloro- | LD50 = 208 mg/kg ( Rat ) | LD50 = 5000 mg/kg ( Rat ) | 170mg/m <sup>3</sup> /4H (Rat) |

**Toxicologically Synergistic** No information available

#### Products

##### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes burns by all exposure routes                                                      |
| Sensitization   | May cause sensitization by skin contact                                                  |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                        | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------------|----------|------------|------------|------------|------------|------------|
| 1,3,5-Triazine, 2,4,6-trichloro- | 108-77-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** Not mutagenic in AMES Test

**Reproductive Effects** No information available.

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>STOT - single exposure</b>                     | Respiratory system                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Symptoms / effects, both acute and delayed</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |

**Endocrine Disruptor Information**

|                              |                                                                |
|------------------------------|----------------------------------------------------------------|
| <b>Other Adverse Effects</b> | The toxicological properties have not been fully investigated. |
|------------------------------|----------------------------------------------------------------|

## 12. Ecological information

**Ecotoxicity**

Do not empty into drains. Reacts with water so no ecotoxicity data for the substance is available.

|                                      |                                            |
|--------------------------------------|--------------------------------------------|
| <b>Persistence and Degradability</b> | No information available                   |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                  |
| <b>Mobility</b>                      | . Is not likely mobile in the environment. |

| Component                        | log Pow |
|----------------------------------|---------|
| 1,3,5-Triazine, 2,4,6-trichloro- | 0.512   |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

**DOT**

|                             |                                      |
|-----------------------------|--------------------------------------|
| <b>UN-No</b>                | UN2670                               |
| <b>Proper Shipping Name</b> | consumer commodity CYANURIC CHLORIDE |
| <b>Hazard Class</b>         | 8                                    |
| <b>Packing Group</b>        | II                                   |

**TDG**

|                             |                   |
|-----------------------------|-------------------|
| <b>UN-No</b>                | UN2670            |
| <b>Proper Shipping Name</b> | CYANURIC CHLORIDE |
| <b>Hazard Class</b>         | 8                 |
| <b>Packing Group</b>        | II                |

**IATA**

|                             |                   |
|-----------------------------|-------------------|
| <b>UN-No</b>                | UN2670            |
| <b>Proper Shipping Name</b> | Cyanuric chloride |
| <b>Hazard Class</b>         | 8                 |
| <b>Packing Group</b>        | II                |

**IMDG/IMO**

|                             |                   |
|-----------------------------|-------------------|
| <b>UN-No</b>                | UN2670            |
| <b>Proper Shipping Name</b> | Cyanuric chloride |
| <b>Hazard Class</b>         | 8                 |
| <b>Packing Group</b>        | II                |

## 15. Regulatory information

### International Inventories

| Component                        | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|----------------------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 1,3,5-Triazine, 2,4,6-trichloro- | 108-77-0 | X   | -    | X    | ACTIVE                                        | 203-614-9 | -      | -   |

| Component                        | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|----------------------------------|----------|-------|----------|------|------|------|------|-------|-------|
| 1,3,5-Triazine, 2,4,6-trichloro- | 108-77-0 | X     | KE-34100 | X    | X    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component                        | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|----------------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| 1,3,5-Triazine, 2,4,6-trichloro- | Part 4 Substance                                     |                                                                            |                                           |

**Legend**

NPRI - National Pollutant Release Inventory

### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component                        | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1,3,5-Triazine, 2,4,6-trichloro- | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

**REACH links**
<https://echa.europa.eu/substances-restricted-under-reach>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                        | CAS-No   | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------------|----------|----------|------------------------------|---------------------------|--------------------------------------------|
| 1,3,5-Triazine, 2,4,6-trichloro- | 108-77-0 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component                        | CAS-No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 1,3,5-Triazine, 2,4,6-trichloro- | 108-77-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

|                         |                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------|
| <b>Prepared By</b>      | Product Safety Department<br>Email: chem.techinfo@thermofisher.com<br>www.thermofisher.com |
| <b>Creation Date</b>    | 22-September-2009                                                                          |
| <b>Revision Date</b>    | 27-March-2024                                                                              |
| <b>Print Date</b>       | 27-March-2024                                                                              |
| <b>Revision Summary</b> | New emergency telephone response service provider.                                         |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**